

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – (Supplementary) 2023 Course: B. Tech. Branch : B. Tech (Common to All) Semester : II Subject Code & Name: Engineering Mathematics-II (BTBS 201) Max Marks: 60 Date:15.01.2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find all the values of $(i)^{\frac{1}{4}}$	Understand (CO1)	6
B)	If $\tan(\theta + i\phi) = \cos \alpha + i \sin \alpha$, prove that $\phi = \frac{1}{2} \log_e \tan \left(\frac{\pi}{4} + \frac{\alpha}{2} \right)$	Understand (CO1)	6
C)	If $p \log(a + ib) = (x + iy) \log m$, prove that $\frac{y}{x} = \frac{2 \tan^{-1} \frac{b}{a}}{\log(a^2 + b^2)}$.	Understand (CO1)	6
Q.2	Solve Any Two of the following.		12
A)	Solve: $\cos^2 x \frac{dy}{dx} + y = \tan x$	Understand (CO2)	6
B)	Solve: $(x^2 y - 2xy^2)dx - (x^3 - 3x^2 y) = 0$	Understand (CO2)	6
C)	A coil having resistance of 20Ω & inductance of $10H$ is connected to $100V$ supply. Determine the values of current after two seconds	Application (CO2)	6
Q. 3	Solve Any Two of the following.		12
A)	Solve: $y'' + 4y' + 13y = 18e^{-2x}$	Understand (CO2)	6
B)	Solve: $(D^2 + 2D + 1)y = e^{-x} \log x$ by method of variation of parameters	Understand (CO2)	6
C)	Solve: $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \cos(\log x)$	Understand (CO2)	6
Q.4	Solve Any Two of the following.		12
A)	Obtain the Fourier series to represent $f(x) = \frac{1}{4}(\pi - x)^2$ in the range $(0, 2\pi)$.	Understand (CO3)	6
B)	Find Fourier series for $f(x) = 9 - x^2$ in the range $(-3, 3)$.	Understand (CO3)	6
C)	Find half range Fourier cosine series for $f(x) = x$, $0 < x < 2$.	Understand (CO3)	6

Q. 5	Solve Any Two of the following.		12
A)	Define divergence and Curl of the vector point function. Find <i>div</i> $\vec{F}$ and <i>Curl</i> $\vec{F}$ if $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$	Understand (CO4)	6
B)	Verify Green's theorem $\int_C [(x^2 - 2xy)dx + (x^2y + 3)dy]$, where C is the boundary of the region bounded by the parabola $y = x^2$ and $y = x$	Understand (CO5)	6
C)	Verify Stokes' theorem for $\vec{F} = xy^2\hat{i} + y\hat{j} + z^2x\hat{k}$ for the surface of the rectangle bounded by $x = 0, y = 0, x = 1, y = 2, z = 0$	Understand (CO5)	6
	*** End ***		